

MULTI-ARMED BANDITS

a) In multi-armed bandits, what is the cardinality of the state space $|S|$?

The space cardinality can be considered to be equal to zero (the concept itself of state doesn't really fit the multi-armed bandits settings). We can also consider such cardinality equals to one as the agent always return to the same state after taking one action (moving to next episode).

b) In RL approaches to solve the multi-armed bandits, how do we estimate the reward distributions R from the experience collecting real world data?

In RL problems (even in the simple multi-armed bandits), the approaches do not aim to estimate a reward distribution, but instead to derive optimal policies or quantities (such as V or Q) that allow us to derive optimal policies or to evaluate the goodness of a given policy.

c) List all of the approaches we have considered to deal with the exploration-exploitation trade-off in the multi-armed bandits problem. What are the hyperparameters of such approaches? Show such hyperparameters by using equations if helpful and illustrate what will happen if we make such hyperparameters bigger.

See lecture 2 and 3 for the related approaches.

- *Epsilon-greedy: the bigger epsilon is, the more we foster exploration (and therefore sacrifice exploitation)*
- *Greedy with Optimistic initialization: the bigger the initial value of Q is, the more we foster exploration (especially in the first steps)*
- *Upper confidence bound: the bigger c , the more we foster exploration*
- *Gradient Bandit Algorithm: the higher alpha, the more we foster exploration*

DYNAMIC PROGRAMMING

a) What are the main motivations to use Dynamic programming approaches?

See lecture 4 and 5.

b) What are the main characteristic of a problem that make it feasible to be tackled with DP?

See lecture 4.

c) Provide the pseudo-code for Policy Iteration ('in place' update version). Which mathematical tools are you using in this algorithm?

See lecture 4.

OFF-POLICY

a) In the full RL problem, what are the motivations to use off-policy approaches? What requirements are needed in order to employ off-policy approaches in the full reinforcement learning setting?

For motivations, look at related material in lecture 7. Requirements are:

- *To know the behaviour policy b ;*
- *To have coverage;*
- *To apply Importance Sampling.*

b) Consider a maze, where agents have the possibility to turn left or right during their exploration; an agent A is exploring the maze, by choosing to turn left or right in the maze with uniform probability. Agent A collect a -1 reward for each time he chooses to turn left or right, and a +100 reward for each time that he is exiting the maze (at that point the episode is terminated). An agent B has instead a policy that will make him choose to go right with 100% probability. Agent B collects 0 reward for each choice (left or right) that he makes and a +100 reward when he is exiting the maze (terminal state). We have available several trajectories collected by agent A: can we use such trajectories to learn how good the policy of Agent B is?

No, it cannot be done. With different rewards we are considering different MDPs.

TD-LEARNING

a) Report the pseudo-code and backup diagrams for SARSA and Expected SARSA.

See lectures 9 and 10.

b) What are the main differences of the two approaches?

See lectures 9, 10.

c) Monte Carlo and TD-learning approaches converge to two different solutions, which ones?

See lectures 9.